BurgFlow MS-40D Installation, Operation & Maintenance - Section 7: Troubl

SECTION 7 - TROUBLESHOOTING GUIDE (MS-40D / MS-32C SERIES)

SYMPTOM 7.3: HIGH DISCHARGE TEMPERATURE / THERMAL TRIP

Probable causes and checks, in order:

(a) Downstream cooler or exchanger fouled - verify exchanger approach temperature against clean baseline; backflush if >5 degC above.

(b) Suction strainer differential pressure high - clean basket if dP exceeds 0.4 bar; low suction pressure causes recirculation heating.

(c) Seal flush line (API Plan 11) blocked - confirm flow at rotameter; inspect flush orifice for scale. A blocked flush line overheats the seal chamber and will present as rising stuffing-box temperature.

(d) Minimum-flow operation - confirm flow instrument reading exceeds the minimum continuous stable flow on the pump datasheet.

SYMPTOM 7.5: PREMATURE MECHANICAL SEAL FAILURE

Abrasive solids in pumped liquid are the leading cause on cooling water duty. Verify strainer condition and water turbidity FIRST. Face scoring with thermal crazing indicates dry running after cavitation - inspect suction line and strainer before blaming the seal.

WARNING: Do not restart after a thermal trip until the cause of restricted cooling flow has been identified and corrected.